

ANDRES SALAZAR

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SUMMARY

Mechanical Engineer with hands-on experience in industrial design, manufacturing, mechanical product development, CAD, FEA, GD&T, prototyping, and field project coordination. Experienced translating production and field requirements into manufacturable designs, technical drawings, and process improvements.

EDUCATION

University of Houston

Bachelor of Science in Mechanical Engineering

Minor in Mathematics

Cumulative College GPA: 3.62 | Magna Cum Laude

Houston, TX

December 2025

SKILLS

Design & Analysis: SolidWorks, Fusion 360, GD&T, Mechanical Drawings, FEA, ANSYS, COMSOL, MATLAB

Manufacturing: CNC/CAM, laser cutting, press brake, 3D printing, prototyping, G-code, design for manufacturability.

Certifications: SolidWorks Professional, Six Sigma Green Belt, Fusion 360, Excel - Copilot Professional

Languages: Spanish (fluent) and English (fluent)

RELEVANT WORK EXPERIENCE

Industrial Design Engineer – Texas Metal Works

Conroe, TX

Industrial Design / Manufacturing

September 2026 – Present

- Develop and revise CAD models, dimensions, and manufacturing drawings for fabricated metal components and production equipment supporting laser cutting, press brake, welding, and assembly operations.
- Evaluate production equipment and quality-control opportunities, including dimensional and flatness verification, to improve repeatability and manufacturability.

Project Supervisor - DH Contractors LLC

Richmond, VA

Project Management Apprenticeship

January 2026 – June 2026

- Support underground fiber and utility installation projects by reviewing construction plans, ROW notes, route maps, pedestal locations, handholes, splice points, and field layout requirements.
- Assist with crew coordination, NC811 utility locates workflows, material tracking, equipment planning, and field issue resolution during HDD, plowing, trenching, and conduit installation.

Engineering Practicum – University of Houston

Houston, TX

Teacher Assistant – Mechanical & Aerospace Engineering Department

May 2024 – December 2025

- Spearhead the development of a hands-on practicum course for 200+ students, equipping freshman and sophomore with industry-standard skills in CAD, 3D Printing, CNC Milling, and Laser Cutting while following safety protocols.
- Design structured, real-world-driven curriculum, creating 50+ CAD models, detailed lectures, quizzes, and assignments to enhance engagement and technical proficiency.

Workshop Facilitator – University of Houston

Houston, TX

Teacher Assistant – Mechanical & Aerospace Engineering Department

January 2024 – May 2024

- Mentored practical preparation for SolidWorks certification, resulting in 100+ students earning CSWA credentials and raising the certification rate from under 30% to over 80%.

Unmanned Aerosol System – Snow College

Ephraim, UT

Research Assistant – Engineering Department

October 2022 – March 2023

- Built a UAS platform to profile vertical temperature, relative humidity, pressure, and particulates, integrating an iMET XQ-2 reference with low-cost Arduino/BME280 and SPS30 sensors.
- Validated sensors via Part-107 flights, generating profiles to ~140 m (400 ft) AGL; confirmed iMET XQ-2's faster temperature response, while assessing durability and accuracy limitations of the BME280.

RELEVANT PROJECTS

Lunar Dust Tolerant Electrical Connector – NASA JSC Power and Propulsion Division

Houston, TX

University of Houston | Capstone Design Project

January 2025-December 2025

- Designed and prototyped a dust-sealed electrical connector for lunar EVA applications, ensuring IP6X-rated protection against abrasive lunar regolith with an innovative gate-sealing mechanism.
- Engineered a one-handed, EVA-glove-compatible operation with integration with robotic systems.
- Conducted FEA (ANSYS) and thermal simulations, including fatigue and vibration analysis, to validate structural integrity and material expansion from -170°C to +127°C in a vacuum environment.
- Collaborated with NASA mentors and suppliers to resolve manufacturability and tolerance challenges, ensuring readiness for prototype fabrication.

Honors & Awards

- American Bureau of Shipping academic excellence scholarship
- Shell Engineering Award

October 2025

February 2024

Commented [AS1]: Task are still in progress. Update once operation improvement happens and specify within x% to y%.